

SCIE1500 Sample Final Exam — Semester 2, 2026

ANSWER KEY

Answer key for the **Sample Final Exam** (the first of the two sample papers). Part I: 36 MCQs with brief justifications (72 marks). Part II: 7 Python MCQs with justifications (14 marks). Part III: *answer keys only* for the four long-answer questions — the final results, not full worked solutions, so that you still have to produce the working yourself. Candidates answer **any two** of the four (24 marks). Total: **110 marks**.

Part I — MCQ Answer Key

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
(b)	(a)	(d)	(a)	(c)	(c)	(d)	(a)	(a)	(a)	(c)	(b)
Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24
(b)	(a)	(a)	(a)	(b)	(d)	(a)	(e)	(c)	(b)	(b)	(b)
Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36
(b)	(d)	(d)	(a)	(a)	(a)	(c)	(a)	(b)	(a)	(a)	(a)

Part I — Brief Justifications

Week 1 — Functions

Q1. Answer: (b) The expression $\sqrt{3x-6}$ requires $3x-6 \geq 0$, i.e. $x \geq 2$. So $D = [2, \infty)$, not $\mathbb{R}$.

Q2. Answer: (a) $f(g(x)) = -e^{-3(-x^2/3)+2} = -e^{x^2+2}$.

Q3. Answer: (d) Complete the square: $y = -(x-3)^2 + 4$. The parabola opens *downward*, so the range is $y \leq 4$; statement (d) claims $y \geq 4$ and is therefore the false one.

Week 2 — Exponentials & Logarithms

Q4. Answer: (a) $P(0) = 3$ and $P(20) = 3e^{1.6} \approx 14.85$. Only plot (a) starts at $P = 3$ and rises exponentially; (b) is linear and decreasing, (c) starts at 30 and decays.

Q5. Answer: (c) Doubling in 2.5 years means $2 = e^{2.5r}$, so $r = \frac{\ln 2}{2.5} \approx 0.277 = 27.7\%$ per year. Option (d) is $\ln 2$ itself — the rate you would get for doubling in one year.

Week 3 — Logistic Growth & Schaefer

Q6. Answer: (c) At $P = K/2$: $1 + Ae^{-\alpha t} = 2 \Rightarrow e^{-\alpha t} = 1/A$, so $t = \frac{\ln A}{\alpha} = \frac{\ln 80}{0.20} \approx 21.9 \approx 22$.

Q7. Answer: (d) $G''(S) = -\frac{2g}{K} < 0$, so G is concave *down*, not concave up. The other three statements are true.

Q8. Answer: (a) Solve $y = 5 + x/3$ for x : $x = 3(y - 5) = 3y - 15$; swapping the variables gives $f^{-1}(x) = -15 + 3x$. Option (d) has the sign of the constant wrong.

Week 4 — Limits & Derivatives

Q9. Answer: (a) $y(3) = e^0 + 2 = 3$ and $y'(x) = e^{x-3}$, so $y'(3) = 1$. The tangent is $y - 3 = 1 \cdot (x - 3)$, i.e. $y = x$.

Q10. Answer: (a) Power rule term by term, with $\frac{d}{dx} \ln x = \frac{1}{x}$: $y' = 2x^3 + x^2 - 2 + \frac{1}{x}$.

Week 5 — Differentiation Rules & Optimisation

Q11. Answer: (c) Product rule: $y' = 2x e^{-x} + x^2(-e^{-x}) = e^{-x}(2x - x^2) = x e^{-x}(2 - x)$. Option (a) differentiates only the first factor.

Q12. Answer: (b) Chain rule (outside $\times$ inside): $y' = 4(3x^2 + 1)^3 \cdot 6x = 24x(3x^2 + 1)^3$. Option (a) omits the inside derivative.

Q13. Answer: (b) $f'(x) = -4x + 12 = 0 \Rightarrow x^* = 3$, and $f(3) = -18 + 36 + 5 = 23$. Since $f'' = -4 < 0$, this is a maximum.

Week 6 — Indefinite Integrals

Q14. Answer: (a) $\int (4x^3 - 2x + 5) dx = x^4 - x^2 + 5x + C$. Option (b) differentiates instead of integrating.

Q15. Answer: (a) $F(x) = x^3 + 2x + C$; the condition $F(1) = 1 + 2 + C = 5$ gives $C = 2$, so $F(x) = x^3 + 2x + 2$. Option (c) substitutes the initial value straight in as the constant.

Week 7 — Definite Integrals & Surplus

Q16. Answer: (a) $\int \ln x dx = x \ln x - x + C$, so from 3 to 15 the area is $(15 \ln 15 - 15) - (3 \ln 3 - 3) = 15 \ln 15 - 3 \ln 3 - 12 \approx 25.32$.

Q17. Answer: (b) Consumer surplus is the area under the demand curve above the equilibrium price line — equivalently $\int_0^{Q^*} D(Q) dQ - P^* Q^*$. Option (a) describes producer surplus; (d) is total revenue.

Week 8 — Lotka-Volterra

Q18. Answer: (d) The conversion efficiency is $\lambda/\beta = 0.0004/0.004 = 0.10 = 10\%$, so (c) is true and (d), claiming 0.04%, is false. Check (b) too: $H^* = \gamma/\lambda = 0.08/0.0004 = 200$ and $P^* = \alpha/\beta = 0.10/0.004 = 25 \checkmark$

Q19. Answer: (a) $P^* = \alpha/\beta$ is linear in α , so doubling α doubles P^* . Note the neat asymmetry: the *prey* equilibrium $H^* = \gamma/\lambda$ does not involve α at all.

Week 9 — Probability

Q20. Answer: (e) Statements (a)–(d) are all true. (e) is false: $P(\text{exactly 3 heads}) = \binom{6}{3}/64 = 20/64 = 5/16$, not $1/2$.

Q21. Answer: (c) Bayes' theorem: $P(T^+) = 0.95(0.01) + 0.02(0.99) = 0.0293$, so $P(D | T^+) = 0.0095/0.0293 \approx 0.32$. The low prevalence, not the test quality, is what keeps this figure small.

Q22. Answer: (b) Addition rule: $P(A \cup B) = 0.4 + 0.3 - 0.1 = 0.6$. Option (a) forgets to subtract the overlap.

Week 10 — Random Variables & Hypothesis Testing

Q23. Answer: (b) $E[X] = 1(0.2) + 2(0.3) + 3(0.3) + 4(0.2) = 2.5$.

Q24. Answer: (b) $P(X \geq 8) = \frac{\binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{45 + 10 + 1}{1024} = \frac{56}{1024}$. Option (a) counts only $P(X = 8)$.

Q25. Answer: (b) The p -value $0.015 < \alpha = 0.05$, so reject H_0 : there is significant evidence the new variant is more infectious. Note (c) is wrong language in any case — we never “accept” a null hypothesis.

Week 11 — Trigonometry

Q26. Answer: (d) $x(t) = 4 \cos(2\pi t)$ has period $T = 2\pi/2\pi = 1$ s and frequency $f = 1/T = 1$ Hz — not 2 Hz, so (d) is the false statement.

Q27. Answer: (d) Amplitude $A = (11 - 3)/2 = 4$; midline $D = (11 + 3)/2 = 7$; period 24 gives $B = 2\pi/24 = \pi/12$. The minimum at $t = 0$ requires $\cos(BC) = -1$, so $BC = \pi$ and $C = \pi/B = 12$. Check: $x(0) = 4 \cos(\pi) + 7 = 3$ ✓ Option (a) has $C = 0$, which puts the *maximum* at midnight.

Q28. Answer: (a) With $\omega = 4\pi$: $T = 2\pi/\omega = 0.5$ s and $f = 1/T = 2$ Hz.

Week 12 — Simultaneous Equations & LP

Q29. Answer: (a) $120 - 3P = -60 + 3P \Rightarrow 6P = 180 \Rightarrow P^* = 30$, and $Q^* = 120 - 90 = 30$.

Q30–Q33 — Mixed-topic extension

Q30. Answer: (a) The denominator vanishes at $x = 2$, giving the vertical asymptote $x = 2$. As $x \rightarrow \pm\infty$, $3/(x - 2) \rightarrow 0$ so $f(x) \rightarrow 5$: horizontal asymptote $y = 5$.

Q31. Answer: (c) $PPV = \frac{0.90 \times 0.05}{0.90 \times 0.05 + 0.05 \times 0.95} = \frac{0.045}{0.0925} \approx 0.49$. Even a good test gives roughly a coin-flip when prevalence is low — the same lesson as Q21.

Q32. Answer: (a) Geometric series with $a = 3$, $r = 2$, $n = 8$: $S_8 = 3 \cdot \frac{2^8 - 1}{2 - 1} = 3 \times 255 = 765$. Option (b) gives only the *final* term $3 \times 2^7 = 384$.

Q33. Answer: (b) Independence means $P(A \cap B) = P(A)P(B) = 0.3 \times 0.4 = 0.12$. Option (a) is $P(A \cup B)$ computed for disjoint events.

Q34–Q36 — Second mixed-topic extension

Q34. Answer: (a) $f''(x) = 6x - 12 = 0 \Rightarrow x = 2$, and $f(2) = 8 - 24 + 5 = -11$. For $x < 2$, $f'' < 0$ (concave down); for $x > 2$, $f'' > 0$ (concave up). Inflection point $(2, -11)$.

Q35. Answer: (a) Chain rule: $x'(t) = 5 \cdot 4\pi \cos(4\pi t) = 20\pi \cos(4\pi t)$. Option (b) forgets the inside derivative 4π .

Q36. Answer: (a) Part 1 of the Fundamental Theorem of Calculus: if $F(x) = \int_a^x f(t) dt$ then $F'(x) = f(x)$ — differentiation undoes integration.

Part II — Python: Answer Key

Q37	Q38	Q39	Q40	Q41	Q42	Q43
(b)	(c)	(b)	(b)	(c)	(b)	(c)

Part II — Justifications

Q37. Answer: (b) Passing the tuple `(x, 0, 20)` to `sp.integrate` makes it a *definite* integral, so CS is the number $\int_0^{20} (100 - 2x - 60) dx$, the exact consumer surplus. Option (c) describes what you would get from `sp.integrate(demand - 60, x)` — the same call *without* the limits.

Q38. Answer: (c) `plt.fill_between(x, y, 0)` shades the region lying between the curve `y` and the constant level 0, i.e. between $y = 3\sin(2x)$ and the x -axis. It is a purely visual command: it shades the area but does not compute or normalise it.

Q39. Answer: (b) `sp.diff(N, t)` differentiates symbolically, giving $\frac{dN}{dt} = 200(0.05)e^{0.05t} = 10e^{0.05t}$. The result is an exact SymPy expression, not a numerical array — arrays come from NumPy.

Q40. Answer: (b) Setting both right-hand sides to zero and solving simultaneously is exactly the definition of an equilibrium. Because the symbols were declared `positive=True`, the trivial solution (0,0) is excluded and what remains is $N^* = \gamma/\delta = 0.1/0.005 = 20$, $P^* = \alpha/\beta = 0.6/0.02 = 30$. A time series (option (d)) would need a numerical integrator such as `odeint`.

Q41. Answer: (c) `binom.cdf(k, n, p)` is the *cumulative* distribution function, $P(X \leq k)$ — so `binom.cdf(8, 12, 0.40)` is $P(X \leq 8)$ and $1 -$ that is $P(X \geq 9)$, the correct one-sided p -value for 9 observed successes. Option (a) describes `binom.pmf`; note also that “density function” in (d) is the wrong term for a discrete variable.

Q42. Answer: (b) The code differentiates, solves $\pi'(t) = 0$ for the stationary point `t_star`, then `.subs(t, t_star)` substitutes it back into the *profit* function. So `max_profit` is the maximum profit value; `t_star` (option (a)) is the separate variable holding the optimal time.

Q43. Answer: (c) At $t = 0$ the exponential is $e^0 = 1$, so $P(0) = \frac{K}{1+A} = \frac{800}{1+99} = 8$ — the initial population. This is the standard reading of A in the logistic model: it is set by how far below K the population starts, not a population itself.

Part III — Answer Key (final answers only)

Candidates answer any **two** of these four; only the two attempted are marked. Listed below are the results you should arrive at — the working is yours to produce.

Q44. Livestock holding period (12 marks)

- (a) $w(t) = 60 + 2.5t - 0.04t^2$ kg per animal.
- (b) $\pi(t) = 28,800 + 1152t - 19.2t^2$ (TR = $480w(t)$, TC = $48t$).
- (c) $t^* = 30$ days; $\pi''(t) = -38.4 < 0$ confirms a maximum.
- (d) $w(30) = 99$ kg/animal; TR = \$47,520; TC = \$1,440; maximum profit = **\$46,080**.

Q45. Diet linear program (12 marks)

- (a) Minimise $C = 3x + 2y$ subject to $5x + 4y \geq 120$, $15x + 20y \geq 500$, $0.5x + y \geq 8$, $x \geq 4$, $y \geq 0$.
- (b) Vertices of the feasible region: (4, 25), (10, 17.5) and (33.33, 0).
- (c) Costs at those vertices: \$62, \$65, \$100. Optimum $(x^*, y^*) = (4, 25)$ at **\$62**.
- (d) Binding: protein (= 120) and the veterinary minimum ($x = 4$). Slack: energy ($560 \geq 500$) and vitamin A ($27 \geq 8$).
- (e) At \$6 per unit of Food I the vertices are unchanged and the optimum stays at (4, 25), new cost **\$74** — it is pinned by the binding $x = 4$ constraint.

Q46. Schaefer fishery (12 marks)

- (a) $S_{\text{MSY}} = K/2 = 400$ tonnes and $\text{MSY} = gK/4 = \mathbf{50}$ tonnes/year.
- (b) At $S = 200$ with $H = 50$: $dS/dt = 37.5 - 50 = -\mathbf{12.5}$ tonnes/year. The stock *declines* — harvesting at MSY while below S_{MSY} drives the fishery towards collapse.
- (c) $H_{\text{max}}(200) = G(200) = \mathbf{37.5}$ tonnes/year.
- (d) Revenue at MSY = $50 \times \$1,000 = \mathbf{\$50,000}$ per year. Operating above MSY pushes S below S_{MSY} , and because $G(S)$ is concave down a lower stock yields a lower sustainable catch — a self-reinforcing decline.

Q47. Drug infusion kinetics (12 marks)

- (a) $\frac{dC}{dt} = \frac{k_{\text{in}}}{V} - k_{\text{out}}C = 0.25 - 0.2C$ (from $dm/dt = k_{\text{in}} - k_{\text{out}}VC$ with $m = VC$).
- (b) $C_{\text{ss}} = 0.25/0.2 = \mathbf{1.25}$ mg/L.
- (c) $C(t) = 1.25(1 - e^{-0.2t})$ mg/L.
- (d) $t_{90} = \frac{\ln 10}{0.2} \approx \mathbf{11.5}$ hours.

— END OF ANSWER KEY —